Peidong **Wang**

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Education

Doctor of Philosophy (Ph.D.) — *Cambridge, MA*

Massachusetts Institute of Technology — *Sep 2019–May 2025*

- Major: Climate Science
- Advisor: Susan Solomon
- Thesis: Understanding Drivers of Stratospheric Ozone Change and Fingerprinting Its Recovery

Bachelor of Science (B.S.) — *Madison, WI*

University of Wisconsin–Madison — *Sep 2015–May 2019*

- Major: Atmospheric & Oceanic Sciences
- Major: Mathematics
- Minor (certificate): Computer Science
- Advisor: Tracey Holloway

Professional Experience

Stanford Science Fellow — *Palo Alto, CA*

Stanford University — *Aug 2025–Present*

Host: Noah Diffenbaugh

Postdoctoral Researcher — *Cambridge, MA*

Massachusetts Institute of Technology — *Jun 2025–Aug 2025*

Host: Susan Solomon

Graduate Research Assistant — *Cambridge, MA*

Massachusetts Institute of Technology — *Sep 2019–May 2025*

Advisor: Susan Solomon

Summer Student Fellow — *Woods Hole, MA*

Woods Hole Oceanographic Institution — *May 2018–Aug 2018*

Advisor: Caroline Ummenhofer

Undergraduate Research Assistant — *Madison, WI*

University of Wisconsin–Madison — *Sep 2016–May 2019*

Advisor: Tracey Holloway

Publications

**Corresponding author*

In Prep / Submitted / Under Review

Wang, P.*, Solomon, S., Deser, C., Thompson, D. W. J., Diffenbaugh, N. S. Antarctic Ozone Depletion Shapes Surface Cooling Pattern and Climate Sensitivity. *Proceedings of the National Academy of Sciences* (under review).

2026

Guan, J.[*], Santer, B. D., **Wang, P.**, Fu, Q., Garcia, R. R., Li, Y., Stone, K., Kinnison, D. E., Zhang, J., Chiodo, G., Lamarque, J.-F., Solomon, S.[*] (2026). The Emergence of Human Influence on the Ozone Layer by the 1960s. *Proceedings of the National Academy of Sciences*, 123(28), e2608286123. HTML | News | Nature Research Highlight

2025

Wang, P.[*], Solomon, S., Scott, J. R., Yvon-Lewis, S. A., Wennberg, P. O., Weiss, R. F., Rigby, M., An, M. (2025). Ocean Outgassing of Methyl Chloroform as an Underestimated Source of Emission. *Geophysical Research Letters*, 52(24), e2025GL118618. HTML

Sweeney, A.[*], Fu, Q.[*], Solomon, S., Po-Chedley, S., Randel, W. J., Steiner, A., Lin, P., Birner, T., Davis, S., **Wang, P.** (2025). Recent Warming of the Southern Hemisphere Subtropical Lower Stratosphere and Antarctic Ozone Healing. *AGU Advances*, 6(4), e2025AV001737. HTML

Wang, P.[*], Solomon, S.[*], Santer, B. D., Kinnison, D. E., Fu, Q., Stone, K. A., Zhang, J., Manney, G. L., Millán, L. F. (2025). Fingerprinting the Recovery of Antarctic Ozone. *Nature*, 639(8055), 646–651. HTML | News

2024

Wang, P.[*], Solomon, S. (2024). Contrasting Chlorine Chemistry on Volcanic and Wildfire Aerosols in the Southern Mid-Latitude Lower Stratosphere. *Geophysical Research Letters*, 51(18), e2024GL110412. HTML

Zhang, J.[*], **Wang, P.**, Kinnison, D., Solomon, S., Guan, J., Stone, K., Zhu, Y. (2024). Stratospheric Chlorine Processing after the Unprecedented Hunga Tonga Eruption. *Geophysical Research Letters*, 51(17), e2024GL108649. HTML

2023

Liu, G.[*], **Wang, P.**, Kwon, Y.-O. (2023). Physical Insights from the Multidecadal Prediction of North Atlantic Sea Surface Temperature Variability Using Explainable Neural Networks. *Geophysical Research Letters*, 50(24), e2023GL106278. HTML

Wang, P.[*], Solomon, S., Lickley, M., Scott, J. R., Weiss, R. F., Prinn, R. G. (2023). On the Influence of Hydroxyl Radical Changes and Ocean Sinks on Estimated HCFC and HFC Emissions and Banks. *Geophysical Research Letters*, 50(18), e2023GL105472. HTML

Wang, P.[*], Solomon, S., Stone, K. (2023). Stratospheric Chlorine Processing after the 2020 Australian Wildfires Derived from Satellite Data. *Proceedings of the National Academy of Sciences*, 120(11), e2213910120. HTML

Solomon, S.[*], Stone, K.[*], Yu, P., Murphy, D. M., Kinnison, D., Ravishankara, A. R., **Wang, P.** (2023). Chlorine Activation and Enhanced Ozone Depletion Induced by Wildfire Aerosol. *Nature*, 615(7951), 259–264. HTML | News

2022

Wang, P.[*], Yuval, J.[*], O'Gorman, P. A. (2022). Non-Local Parameterization of Atmospheric Subgrid Processes with Neural Networks. *Journal of Advances in Modeling Earth Systems*, 14(10), e2022MS002984. HTML

Wang, P., Holloway, T.[*], Bindl, M., Harkey, M., De Smedt, I. (2022). Ambient Formaldehyde over the United States from Ground-Based (AQS) and Satellite (OMI) Observations. *Remote Sensing*, 14(9), 2191. HTML

2021

Liu, G., **Wang, P.**, Beveridge, M., Kwon, Y.-O., Drori, I. (2021). Predicting Atlantic Multidecadal Variability. *NeurIPS 2021 Workshop on Tackling Climate Change with Machine Learning*. [Awarded with Best Paper: Pathway to Impact]. HTML

Lickley, M.*, Solomon, S., Kinnison, D., Krummel, P., Mühle, J., O'Doherty, S., Prinn, R., Rigby, M., Stone, K. A., **Wang, P.**, Weiss, R., Young, D. (2021). Quantifying the Imprints of Stratospheric Contributions to Interhemispheric Differences in Tropospheric CFC-11, CFC-12, and N_2O Abundances. *Geophysical Research Letters*, 48(15), e2021GL093700. HTML

Wang, P.*, Scott, J. R., Solomon, S., Marshall, J., Babbin, A. R., Lickley, M., Thompson, D. W. J., DeVries, T., Liang, Q., Prinn, R. G. (2021). On the Effects of the Ocean on Atmospheric CFC-11 Lifetimes and Emissions. *Proceedings of the National Academy of Sciences*, 118(12), e2021528118. HTML | News

Thesis

Wang, P. (2025). Understanding Drivers of Stratospheric Ozone Change and Fingerprinting Its Recovery. *Massachusetts Institute of Technology*. HTML

Honors & Awards

2025–2028	**Stanford Science Fellow**, Stanford University
2025	**Carl-Gustaf Rossby Award for the Best PhD Thesis**, PAOC, MIT
2024–2025	**Jule Charney Fellowship**, PAOC, MIT
2023	**Chinese Government Award for Outstanding Self-Financed Students Abroad**, with the extraordinary prize for the top 20 awardees
2022	**Early Career Scientist Best Oral Presentation**, 7th SPARC General Assembly
2021–2022	**Norman C. Rasmussen Fellowship**, EAPS, MIT
2020–2021	**John H. Carlson Fellowship**, EAPS, MIT
2019	**Jule Charney Prize**, PAOC, MIT
2019–2020	**MIT Presidential Fellowship**, MIT
2019	**Herfurth-Kubly Awards for Comprehensive Undergraduate Excellence**, UW–Madison
2019	**University Book Store Excellence Award**, UW–Madison
2018	**Lyle Horn Scholarship Award**, AOS, UW–Madison
2018	**Summer Student Fellowship**, Woods Hole Oceanographic Institution
2017–2018	**Wisconsin Hilldale Undergraduate/Faculty Research Fellowship**, UW–Madison

Teaching

12.003 Introduction to Atmosphere, Ocean, and Climate Dynamics *Fall 2021*
Massachusetts Institute of Technology *Teaching Assistant*
Department of Earth, Atmospheric and Planetary Sciences

CS200 Programming I *Fall 2017*
University of Wisconsin–Madison *Teaching Assistant*
Department of Computer Sciences

CS302 Introduction to Programming *Spring 2017*
University of Wisconsin–Madison *Teaching Assistant*
Department of Computer Sciences

Selected Presentations

Invited talks

- The Antarctic Ozone Hole and Global Climate Change. Fudan University, Sep 2026.
- A Two-Way 'Conversation' Between the Stratosphere and the Ocean. Caltech, Jun 2026.
- Antarctic ozone loss shapes surface cooling pattern and climate sensitivity. ECS & cloud feedback virtual symposia, May 2026.
- From Stratospheric Ozone to the Ocean: Signals Behind the Patterns. CU Boulder, Apr 2026.
- Fingerprinting the Recovery of Antarctic Ozone. Peking University, Mar 2025.
- Contrasting stratospheric chlorine processes on volcanic and wildfire aerosols. Jinan University, Jan 2024.

Conference presentations

- JpGU-AGU Joint Meeting (Oral), Chiba, Japan, May 2026.
- AGU Fall Meeting (Oral), New Orleans, Louisiana, U.S., Dec 2025.
- CESM Working Group Meeting (Oral), Boulder, Colorado, U.S., Feb 2025.
- Quadrennial Ozone Symposium (Poster), Boulder, Colorado, U.S., Jul 2024.
- EGU General Assembly (Oral), Vienna, Austria, Apr 2024.
- AGAGE 68th Meeting (Oral), Boston, Massachusetts, Oct 2023.
- VolImpact Summer School (Poster), Greifswald, Germany, Sep 2023.
- SPARC 7th General Assembly (Oral), Boulder, Colorado, Oct 2022.
- NeurIPS 2021 Workshop on Tackling Climate Change with Machine Learning (Spotlight talk), virtual, Dec 2021.
- Quadrennial Ozone Symposium (Poster), online, Oct 2021.
- AGU Fall Meeting (Poster), Washington, D.C., Dec 2018.
- Symposium in Honor of Isaac Held (Poster), Princeton University, NJ, Oct 2018.
- HAQAST 3rd Meeting, Columbia University, NY, Nov 2017.

Academic Service

Committee Member PAOC Colloquium Committee (2021–2025)

Proposal Reviewer NASA ROSES panelist (2024)

Journal Reviewer *Advances in Climate Change Research* · *Communications Earth & Environment* · *Earth's Future* · *Environmental Science & Technology* · *Geophysical Research Letters* · *Geoscience Letters* · *Global and Planetary Change* · *Nature Communications* · *npj Climate and Atmospheric Science* · *Remote Sensing of Environment* · *Science Advances*